Camera Parameters and Calibration

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Sumary

- Camera parameters
  - Pinhole geometry
  - Basic optics
  - Camera model
- Camera calibration
  - Single camera calibration
  - RanSAC
- AruCO markers
- Example and Demos

Camera parameters

Camera parameters are normally divided in two groups:

- **Intrinsic:**
  Parameters necessary to link the image pixel coordinates with the corresponding coordinates in the camera reference frame.

- **Extrinsic:**
  Parameters that define the *location* and *orientation* of the camera reference frame with respect to a known world reference frame.

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Camera parameters

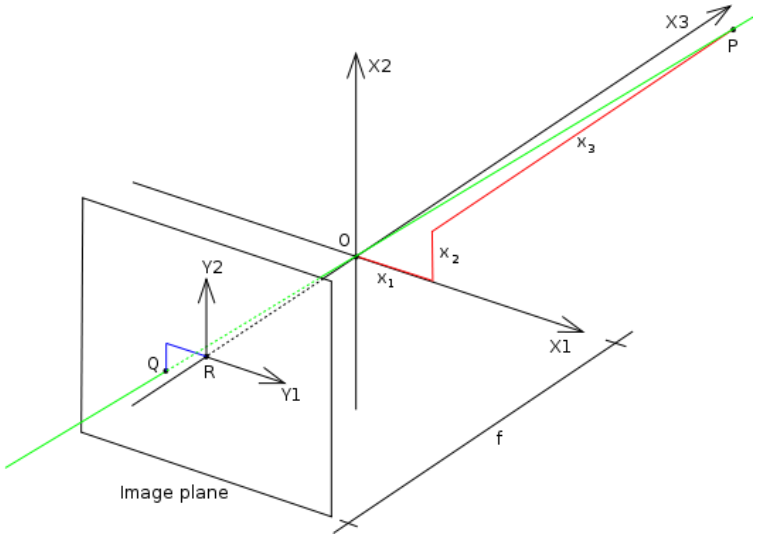
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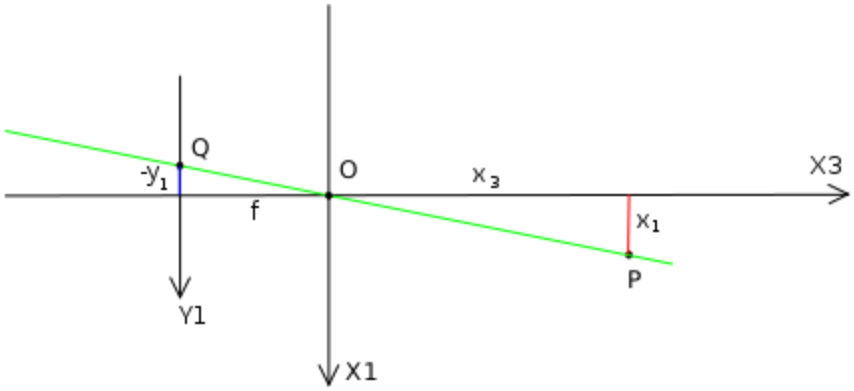
- **Intrinsic:**
  Parameters necessary to link the image pixel coordinates with the corresponding coordinates in the camera reference frame.

Camera model

The pinhole model

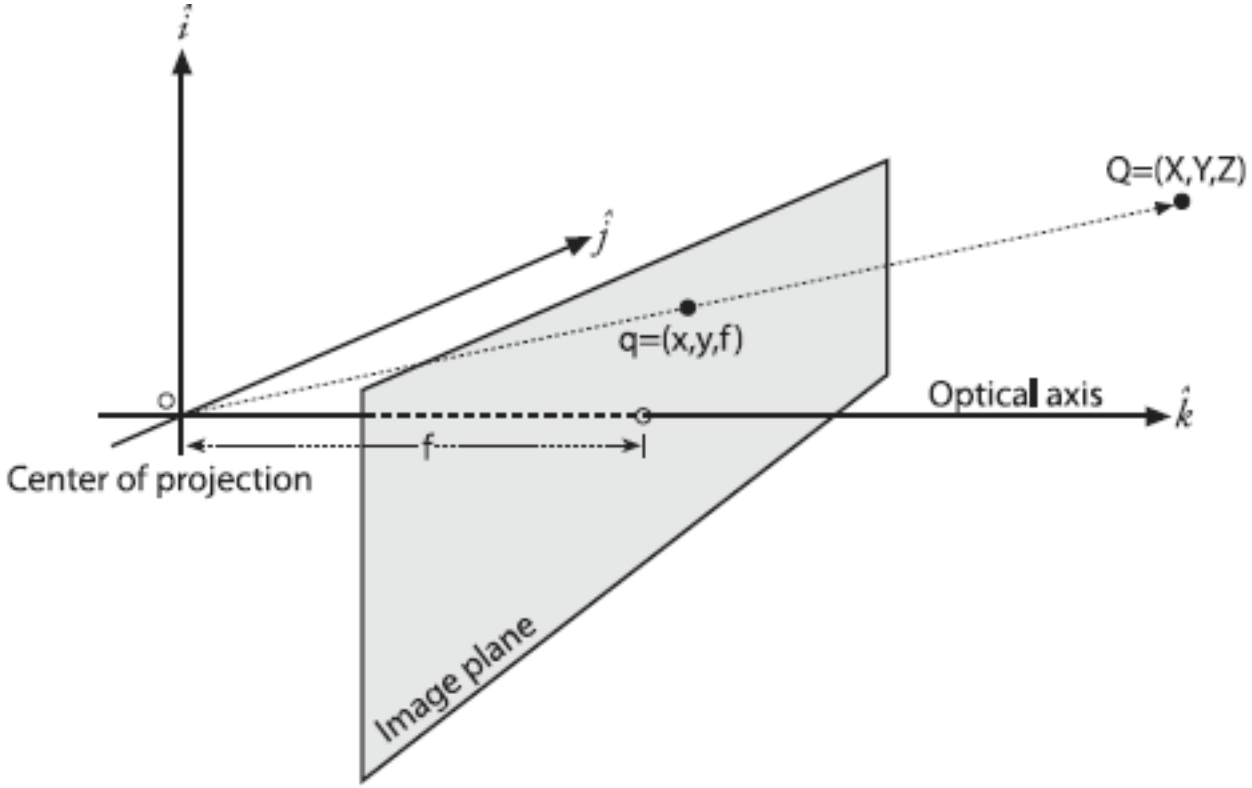
- In most image processing tasks, the optical system can be approximated by the pinhole model.
- The lens is replaced by a narrow opening (pinhole) through which lights go through into the image acquisition plane.
- The point stroked by a light ray going through the pin hole in the direction of the lens main axis is the image plane origin
- In this model the acquisition plane lies at the focal distance from the pinhole.





Camera model

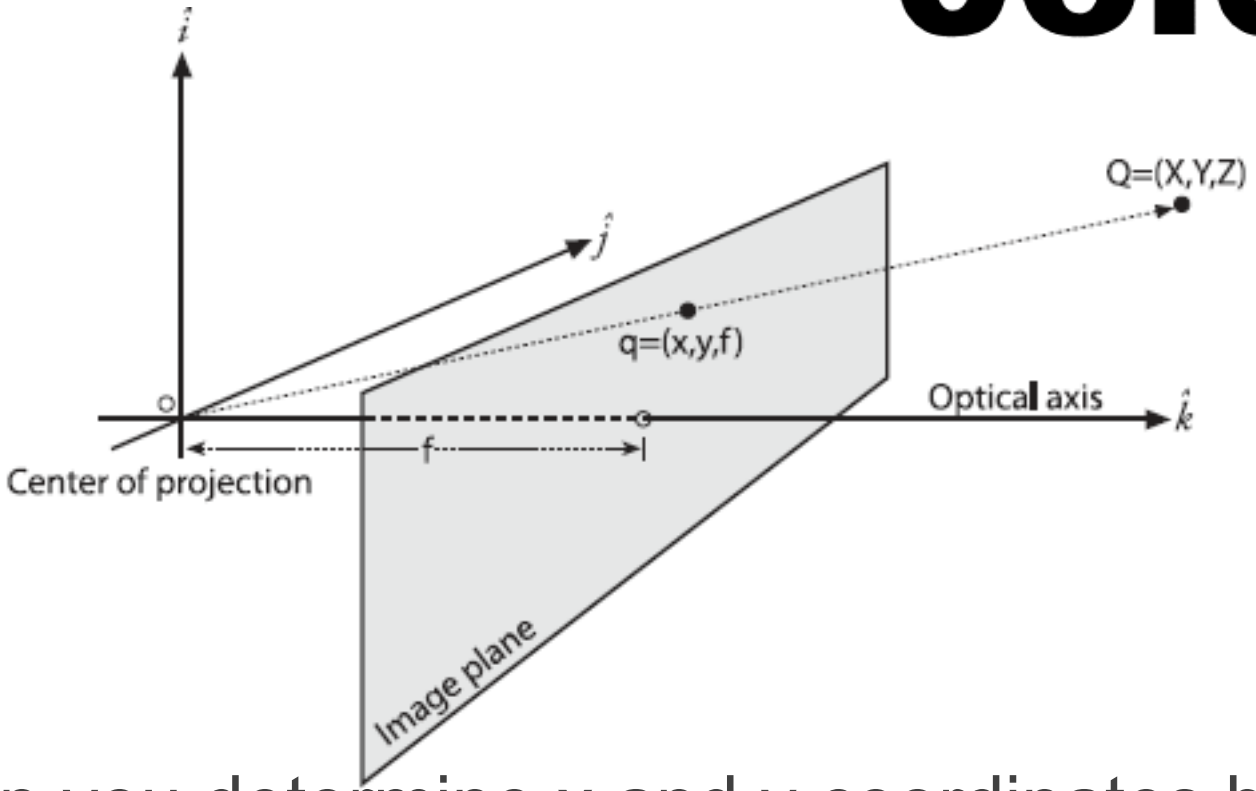
- Pinhole camera model



Camera model

- Pinhole camera model

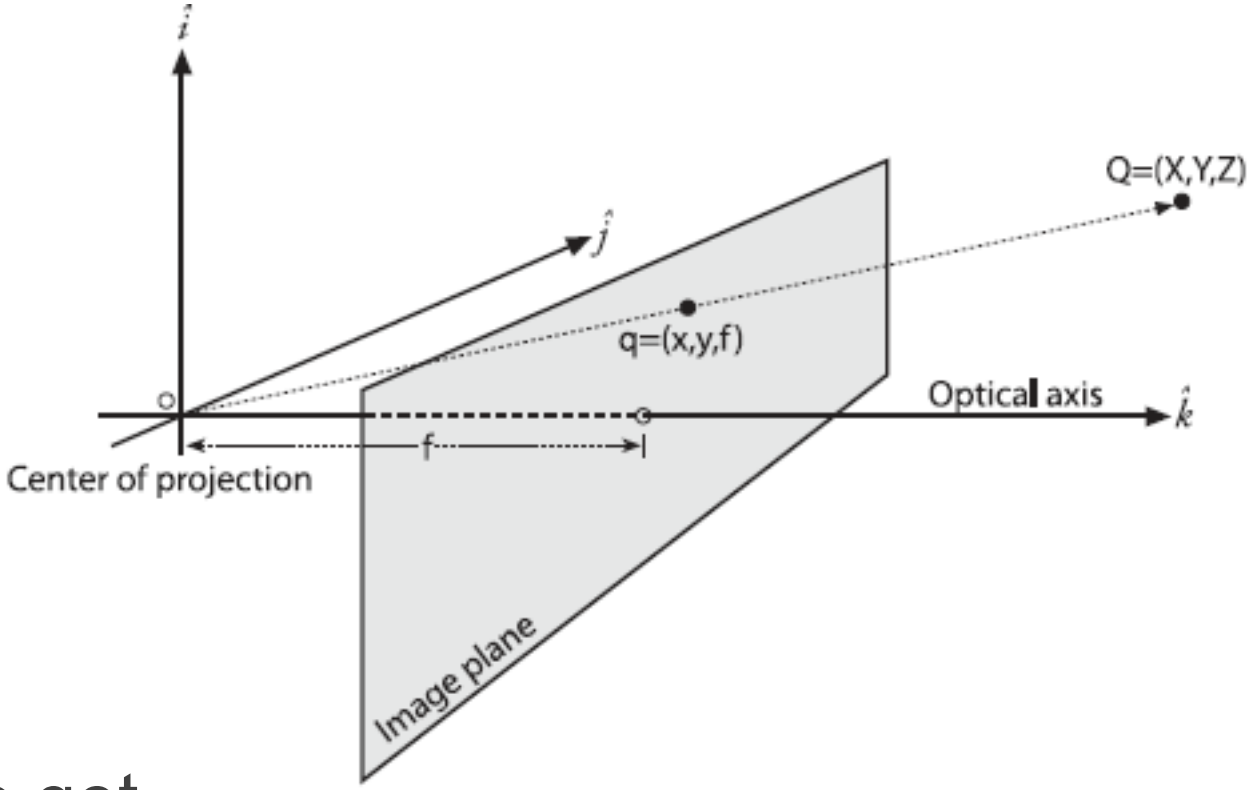
03:00



- Can you determine x and y coordinates based on f, X, Y and Z?

Camera model

- Pinhole camera model



- We get

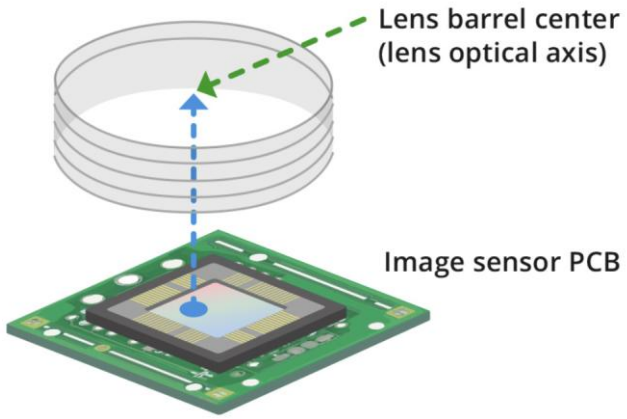
$$x = f\frac{X}{Z} \qquad y = f\frac{Y}{Z}$$

Camera model: optical centre

- Image optical centre and sensor centre are normally not aligned:

$$x_{screen} = f_x \frac{X}{Z} + c_x$$

$$y_{screen} = f_y \frac{Y}{Z} + c_y$$



- f_x e f_y are derived from focal distance but take in consideration pixel size of the sensor (typically rectangular).

https://www.opto-e.com/en/basics/Image-sensor-alignment

Homogeneous coordinates

$$x_{screen} = f_x \frac{X}{Z} + c_x$$

$$y_{screen} = f_y \frac{Y}{Z} + c_y$$

- In homogeneous coordinates, previous equations can be written as
- $q = MQ$

with

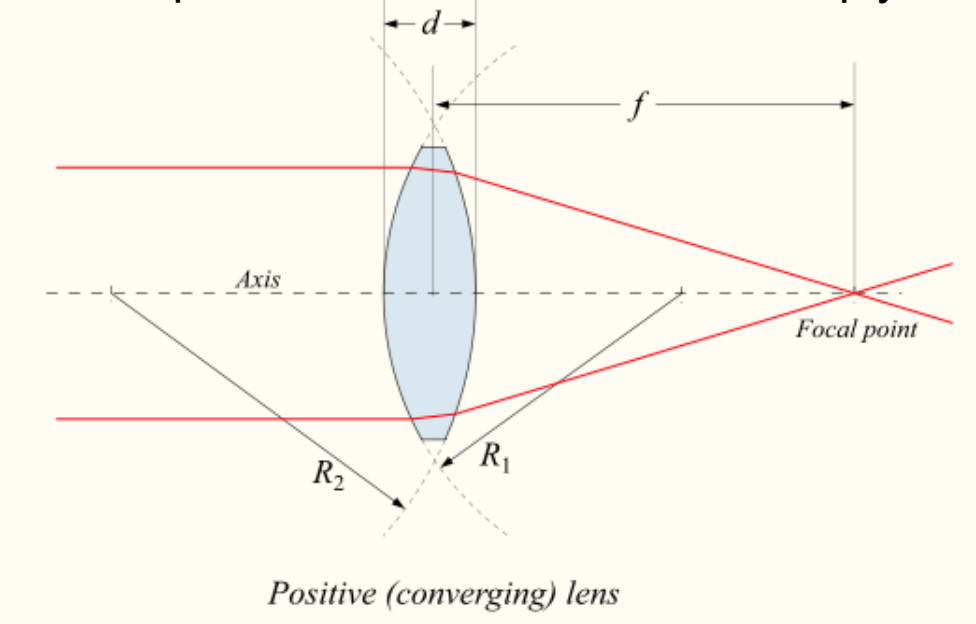
- $q = \begin{bmatrix} x \\ y \\ w \end{bmatrix}, M = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}, Q = \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}$
- Normalizing with w=1, we get the same equations.

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Optical lenses

- One of the most important element of an image acquisition device is its Optical element.
- They can be made of complex groups of lenses (in particular when zooming is desirable).
- Since most simple image acquisition systems, however, use a single converging lens as its optical interface, we will look simply into this case.



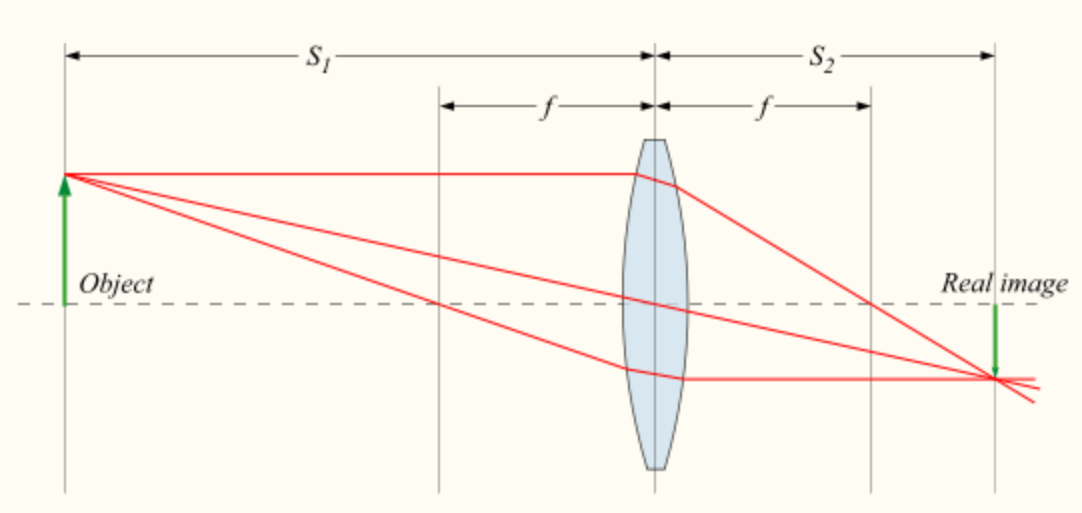
Positive (converging) lens

- Formation of image

- If the distances from the object to the lens and from the lens to the image are S1 and S2 respectively, for a lens of negligible thickness, the distances are related by the thin lens formula which is an acceptable approximation to the full optical equation

$$\frac{1}{S_1} + \frac{1}{S_2} = \frac{1}{f}$$

- If S1 >> S2 then S2 ~= f

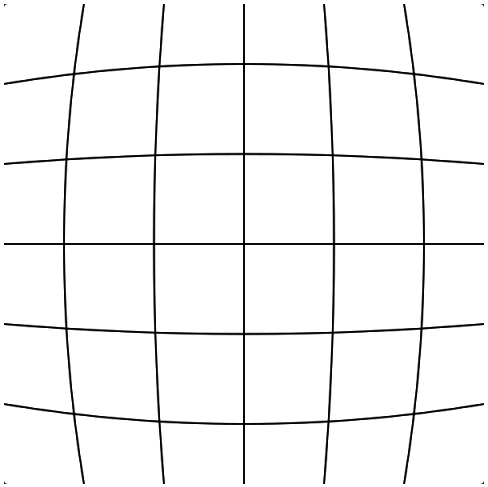


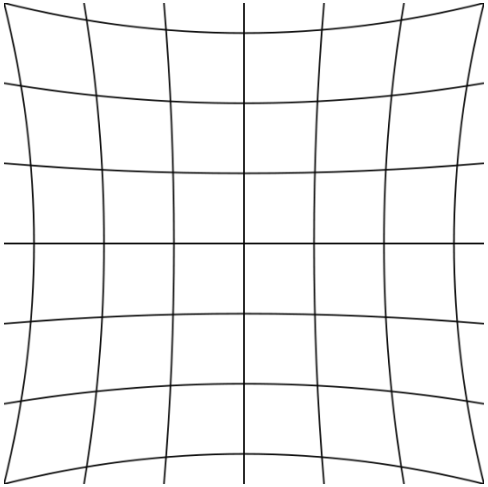
Spherical Aberration

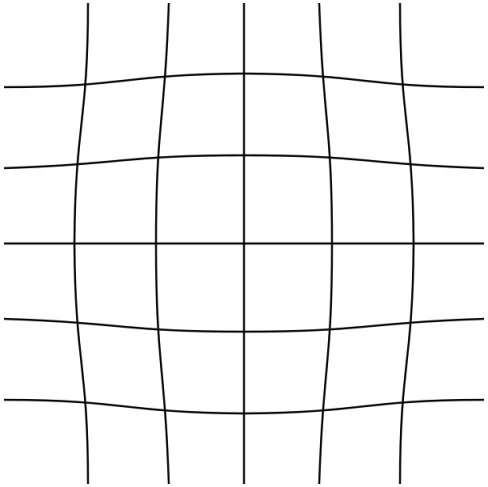
Spherical Aberration is an image deformation resulting from the fact that most lenses have a spherical surface cut which is easier to make than its proper shape.

Spherical Aberration can show as a barrel form (left side), pincushion effect (center) and Mustache distortion (right).

The barrel distortion is by far the most common one and normally increases with diminishing focal distance.







Spherical Aberration

Spherical Aberration results in multiple focus points depending on the distance at which each ray of light enters the lens when referred to its major axis, and can be corrected by can be corrected using the simplified Brown's distortion model

$$x_u = (x_d - x_o)(1 + K_1 r^2 + K_2 r^4 + \ldots)$$
$$y_u = (y_d - y_o)(1 + K_1 r^2 + K_2 r^4 + \ldots)$$

where

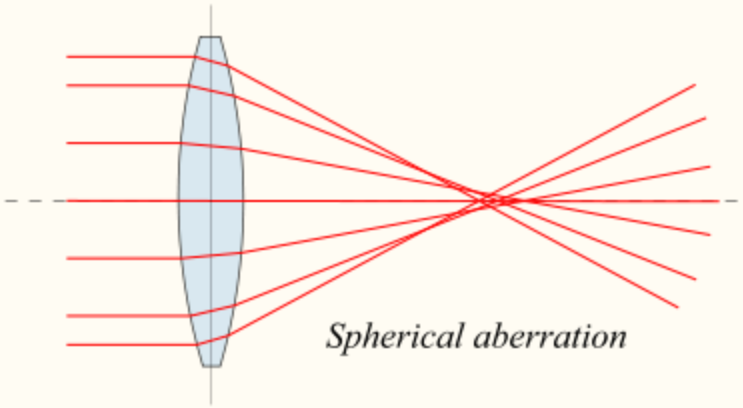
(x_d, y_d) - distorted image point as projected on image plane using specified lens

(x_u, y_u) - undistorted image point as projected by an ideal pin-hole camera

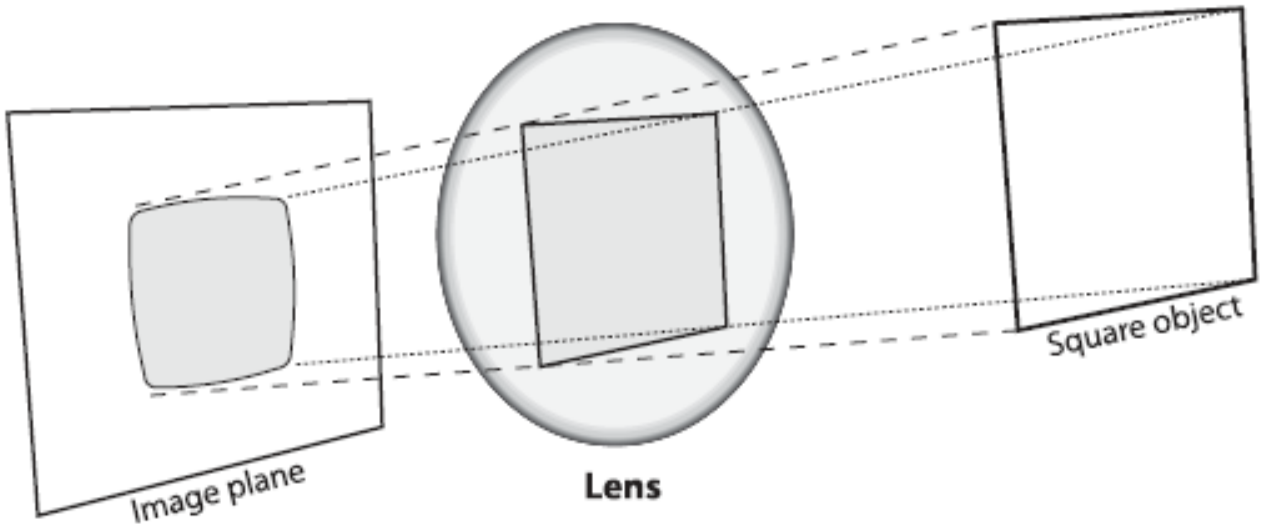
(x_o, y_o) - distortion center (assumed to be the principal point)

$K_n = n^{th}$ - radial distortion coefficient

$$r = \sqrt{(x_d - x_o)^2 + (y_d - y_o)^2}$$



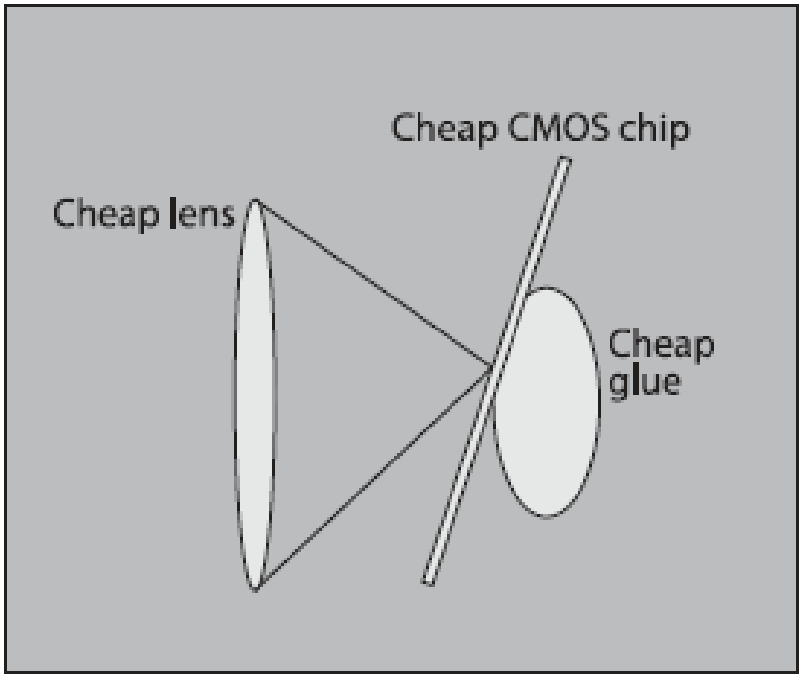
Lens distortion – Radial Distortion

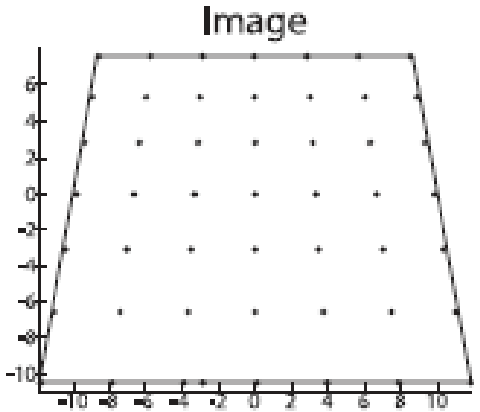


- $x_{corrected} = x(1 + k_1 r^2 + k_2 r^4 + k_3 r^6)$
- $y_{corrected} = y(1 + k_1 r^2 + k_2 r^4 + k_3 r^6)$

Lens distortion – Tangential Distortion

– Tangential





– $x_{corrected} = x + [2p_1 y + p_2(r^2 + 2x^2)]$

– $y_{corrected} = y + [p1(r^2 + 2y^2) + 2p_2 x]$

Camera parameters

Camera parameters are normally divided in two groups:

- **Extrinsic:**
  Parameters that define the *location* and *orientation* of the camera reference frame with respect to a known world reference frame.

Translation and rotation matrices

- Necessary also to find out the Pose (rotation and translation relative to a given coordinate system) of cameras

Geometric Transformations From 2D to 3D

- Most of what we cover before works just the same (see 2D Geometrical Transformations)
- Extend homogenous coordinates notation with one more value
- Transforms, combination, affine have similar properties
- Rotations are the main difference

3D Transformations

- 3D Translation

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & t_x \\ 0 & 1 & 0 & t_y \\ 0 & 0 & 1 & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

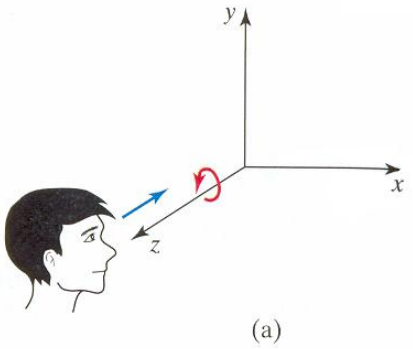
- 3D Scaling

$$\mathbf{S} = \begin{bmatrix} s_x & 0 & 0 & 0 \\ 0 & s_y & 0 & 0 \\ 0 & 0 & s_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

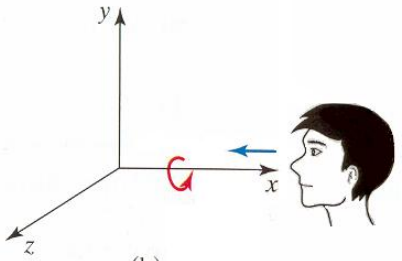
3D Rotation

- Rotation around each one of the coordinate axis

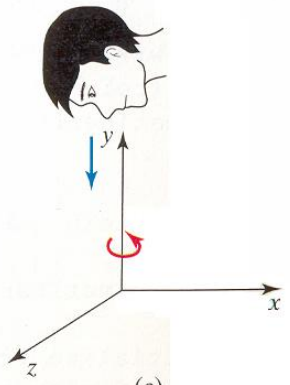
- Positive rotations are CCW (counter clock wise)!!



(a)

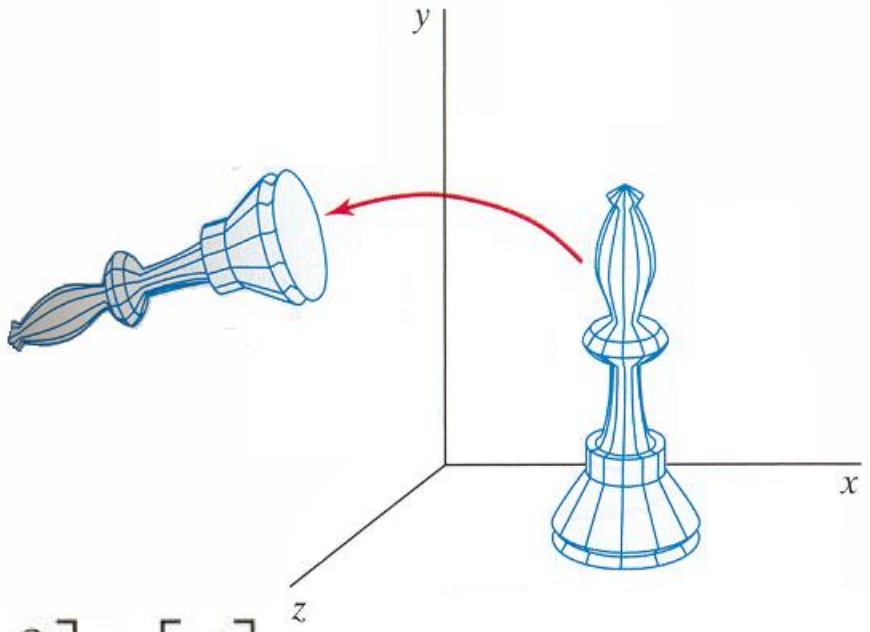


(b)



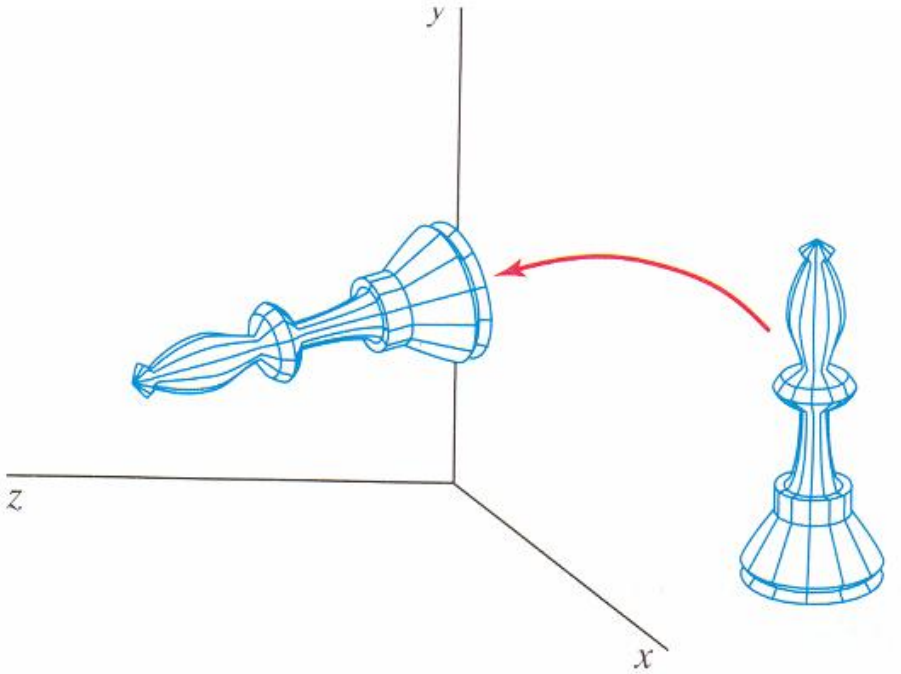
(c)

Rotation around ZZ’



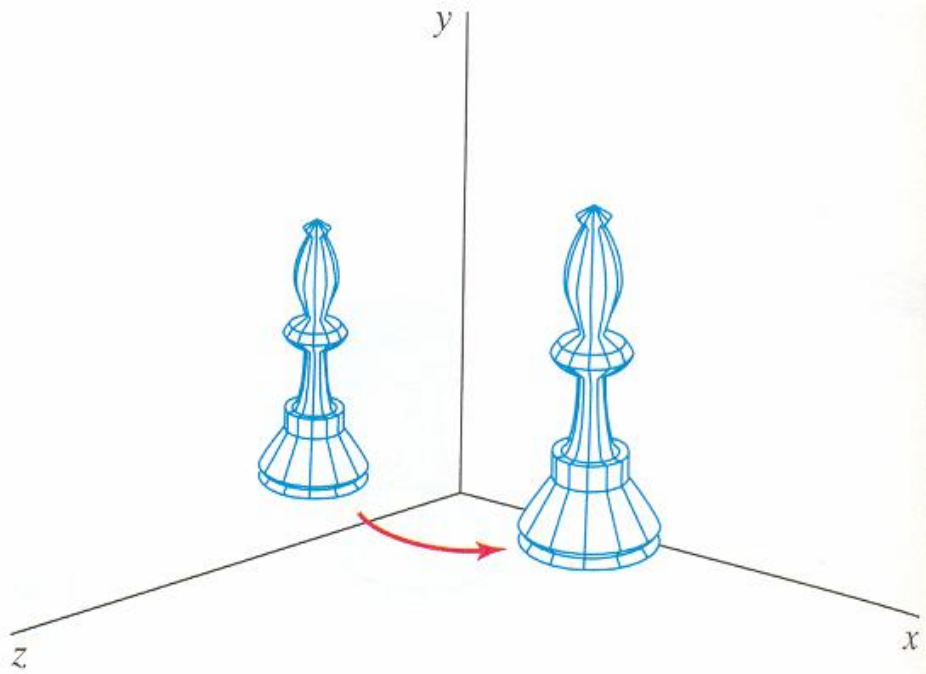
$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 & 0 \\ \sin\theta & \cos\theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

Rotation around XX’



$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta & 0 \\ 0 & \sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

Rotation around YY’



$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & 0 & \sin\theta & 0 \\ 0 & 1 & 0 & 0 \\ -\sin\theta & 0 & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

Rotation Axis

- 2D = rotation is around a point
  - 2D rotation just an angle
- 3D = rotation is around an axis
  - 3D rotation many equivalent representations
    - Axis (direction) and angle
    - Euler angles: 3 angles about 3 axes
    - Quaternions

Homogenous Coordinates – Remember?

Rotation of a vector defined by two points in a Cartesian system can be obtained from:

$$\begin{bmatrix}\mathbf{d}_x\\ \mathbf{d}_y\\ \mathbf{d}_z\end{bmatrix} = \begin{bmatrix}1 & 0 & 0\\ 0 & \cos\theta_x & -\sin\theta_x\\ 0 & \sin\theta_x & \cos\theta_x\end{bmatrix} \begin{bmatrix}\cos\theta_y & 0 & \sin\theta_y\\ 0 & 1 & 0\\ -\sin\theta_y & 0 & \cos\theta_y\end{bmatrix} \begin{bmatrix}\cos\theta_z & -\sin\theta_z & 0\\ \sin\theta_z & \cos\theta_z & 0\\ 0 & 0 & 1\end{bmatrix} \left(\begin{bmatrix}\mathbf{a}_x\\ \mathbf{a}_y\\ \mathbf{a}_z\end{bmatrix} - \begin{bmatrix}\mathbf{c}_x\\ \mathbf{c}_y\\ \mathbf{c}_z\end{bmatrix} \right)$$

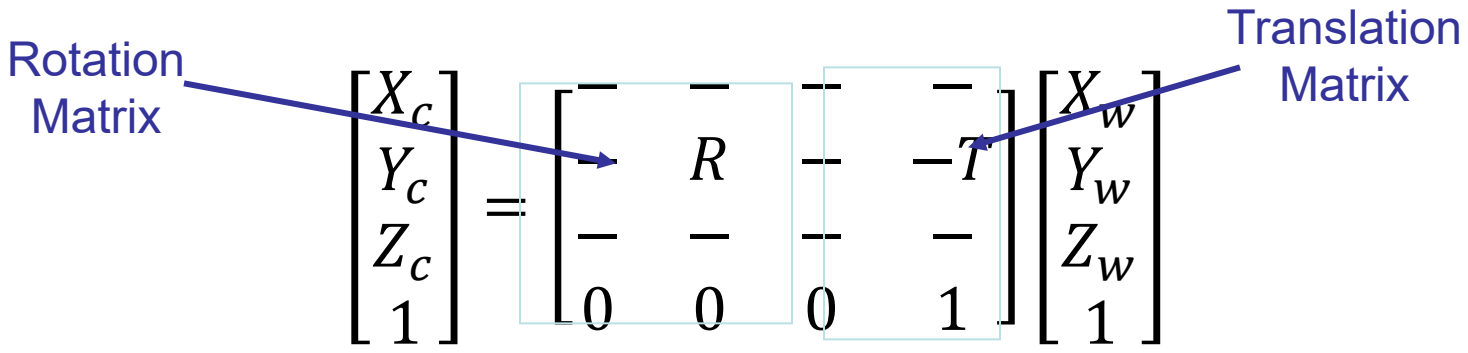
a_n : coordinates of the point to be projected,
c_n : the rotation center coordinates
D_n : resulting rotated vector.

Translation vector ***Tv***, for the point ***p*** can, on the other hand, be defined in a homogenous form by

$$T_{\mathbf{v}}\mathbf{p} = \begin{bmatrix}1 & 0 & 0 & v_x\\ 0 & 1 & 0 & v_y\\ 0 & 0 & 1 & v_z\\ 0 & 0 & 0 & 1\end{bmatrix} \begin{bmatrix}p_x\\ p_y\\ p_z\\ 1\end{bmatrix} = \begin{bmatrix}p_x + v_x\\ p_y + v_y\\ p_z + v_z\\ 1\end{bmatrix} = \mathbf{p} + \mathbf{v}$$

Homogenous Coordinates

One of the most interesting things in the use of homogenous coordinates is that allows to combine a rotation matrix and a translation matrix into a single homogeneous matrix.



Translation and rotation matrices

- Necessary also to find out the Pose (rotation and translation relative to a given coordinate system) of cameras
  - Rotation – vector with 3 rotation angles (r_x, r_y, r_z) that might be composed in a 3x3 rotation matrix.
  - Translation – vector with 3 values (t_x, t_y, t_z)

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Parameters of the general model

Referring to the pinhole camera model, a camera matrix can therefore be used to denote a general projective mapping 3D world coordinates with 2D homogeneous coordinates of pixels in images.

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = A \begin{bmatrix} R & T \end{bmatrix} \begin{bmatrix} x_w \\ y_w \\ z_w \\ 1 \end{bmatrix}$$

Where A represents the camera intrinsic parameters, R and T the rotation and translation matrix x_n each of the coordinates in 3D space and ***u*** and ***v*** the indexes of each pixel on the frame buffer.

$$\mathrm{A} = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}$$

Parameters of the general model

and distortion?

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = A \begin{bmatrix} R & T \end{bmatrix} \begin{bmatrix} x_w \\ y_w \\ z_w \\ 1 \end{bmatrix}$$

OpenCV Camera model

- OpenCV Camera model:
  - 4 intrinsic parameters:
    - Focal distance: f_x, f_y
    - Optical centre: c_x, c_y
  - 5 distortion parameters
    - Lens distortion: k_1, k_2, k_3, p_1, p_2
  - 6 extrinsic parameters:
    - Rotation: r_x, r_y, r_z
    - Translation: t_x, t_y, t_z

Total: **15 parameters**

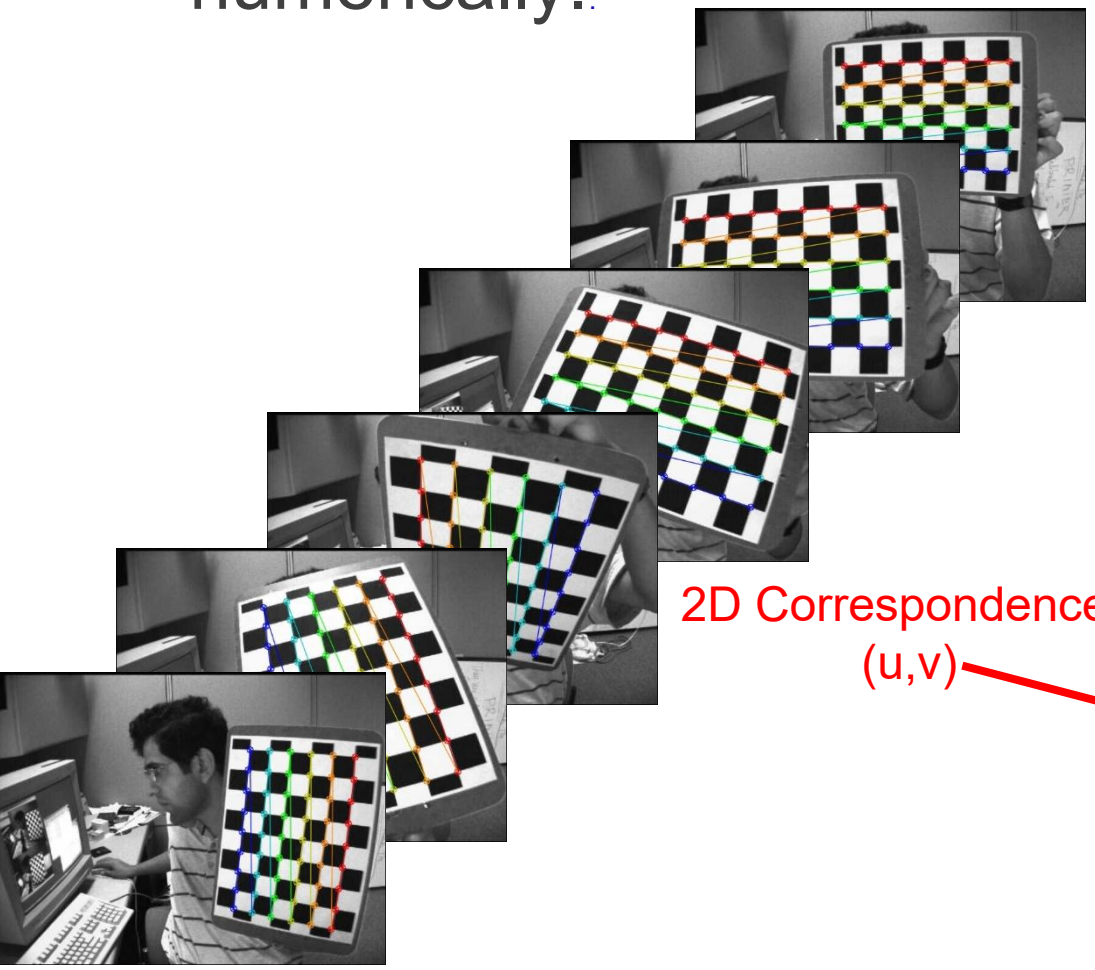
- Other models: Tsai, Heikkila, Zhang

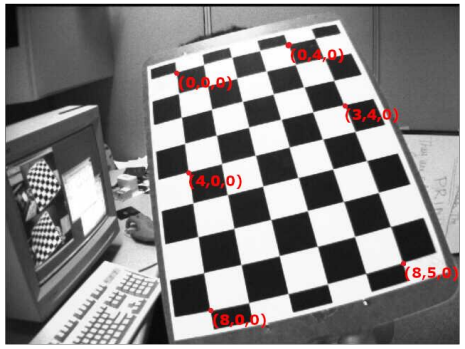
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Single camera calibration

- Given 3D-2D correspondences, possible to define a system of equation that might be solved numerically..





3D Coordinates
(x_w, y_w, z_w)

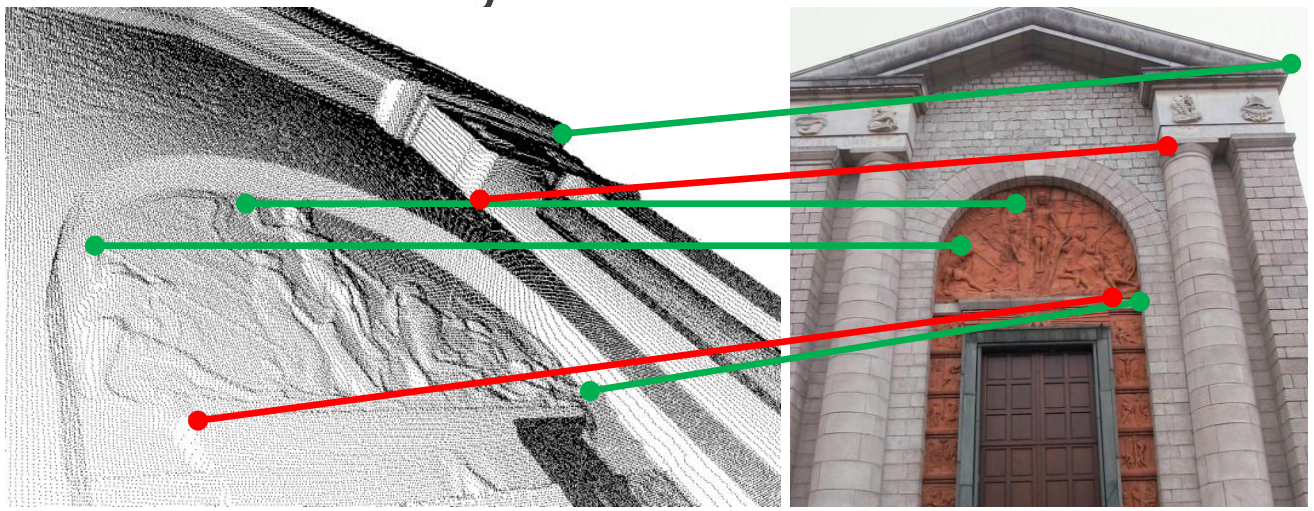
2D Correspondences
(u,v)

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = A \begin{bmatrix} R & T \end{bmatrix} \begin{bmatrix} x_w \\ y_w \\ z_w \\ 1 \end{bmatrix}$$

Single camera calibration

- Given 3D-2D correspondences, possible to define a system of equation that might be solved numerically.

=>Problem: Occurrence of Outliers - Only some matches are mutually consistent



Images from 3 Dimensional Reconstruction Of Real World Scenes Using Laser and Intensity Data. Dias P. PhD Thesis, University of Aveiro, 2003.

- RANSAC - "RANdom SAmple Consensus" might be used to remove outliers

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Flash Search

Please search

RanSAC

and keep the best definition

03:00



Random sample consensus (RANSAC)

- RAndom SAmple Consensus
  - Iterative method to estimate parameters of a mathematical model from a set of observed data that contains outliers
  - Main idea: outlier detection
  - Uses repeated random sub-sampling.

- Idea
  - All the inliers will agree with each other on the camera pose; the (hopefully small) number of outliers will (hopefully) disagree with each other

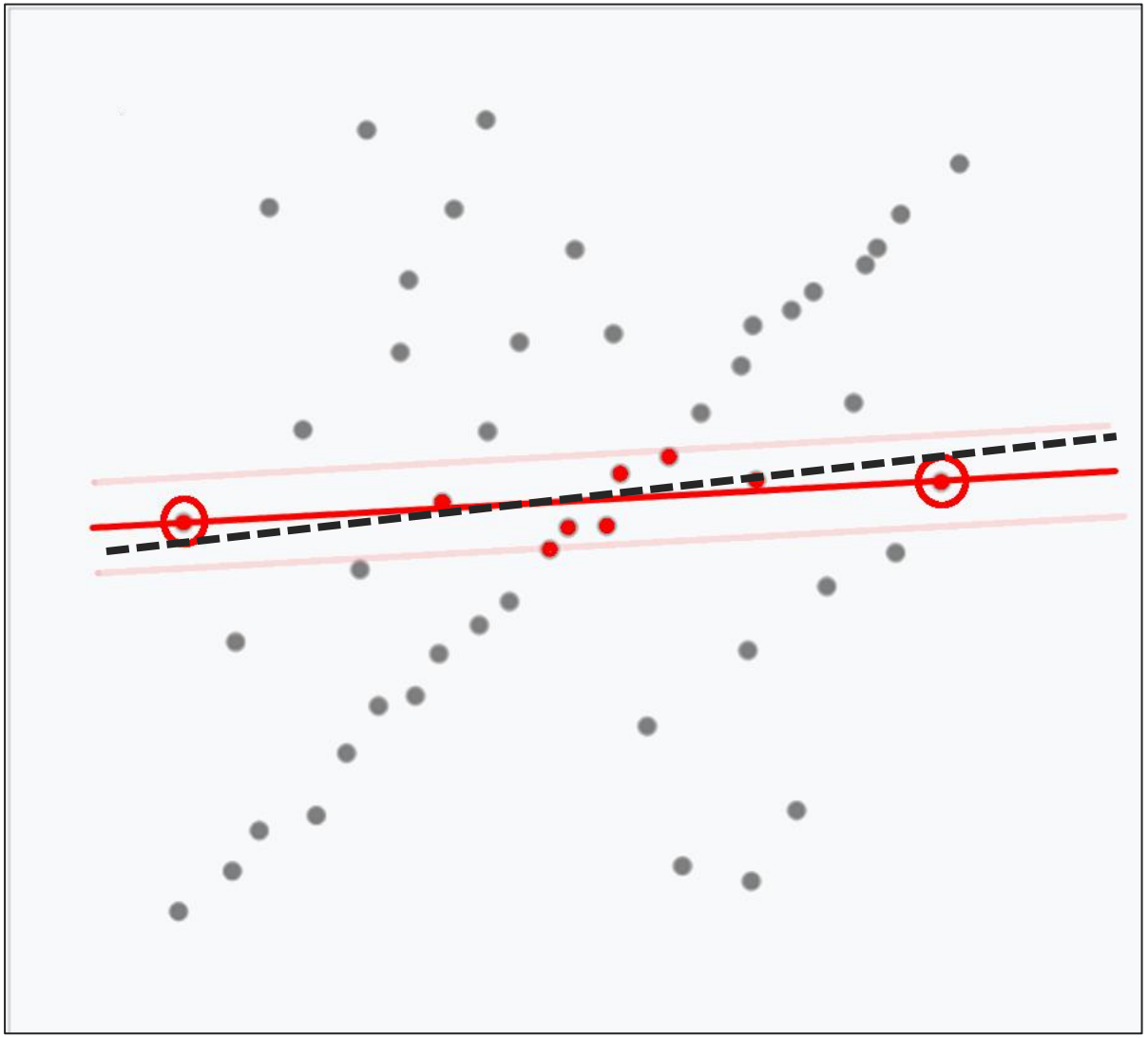
RANSAC – fitting a line

- Dataset with outliers



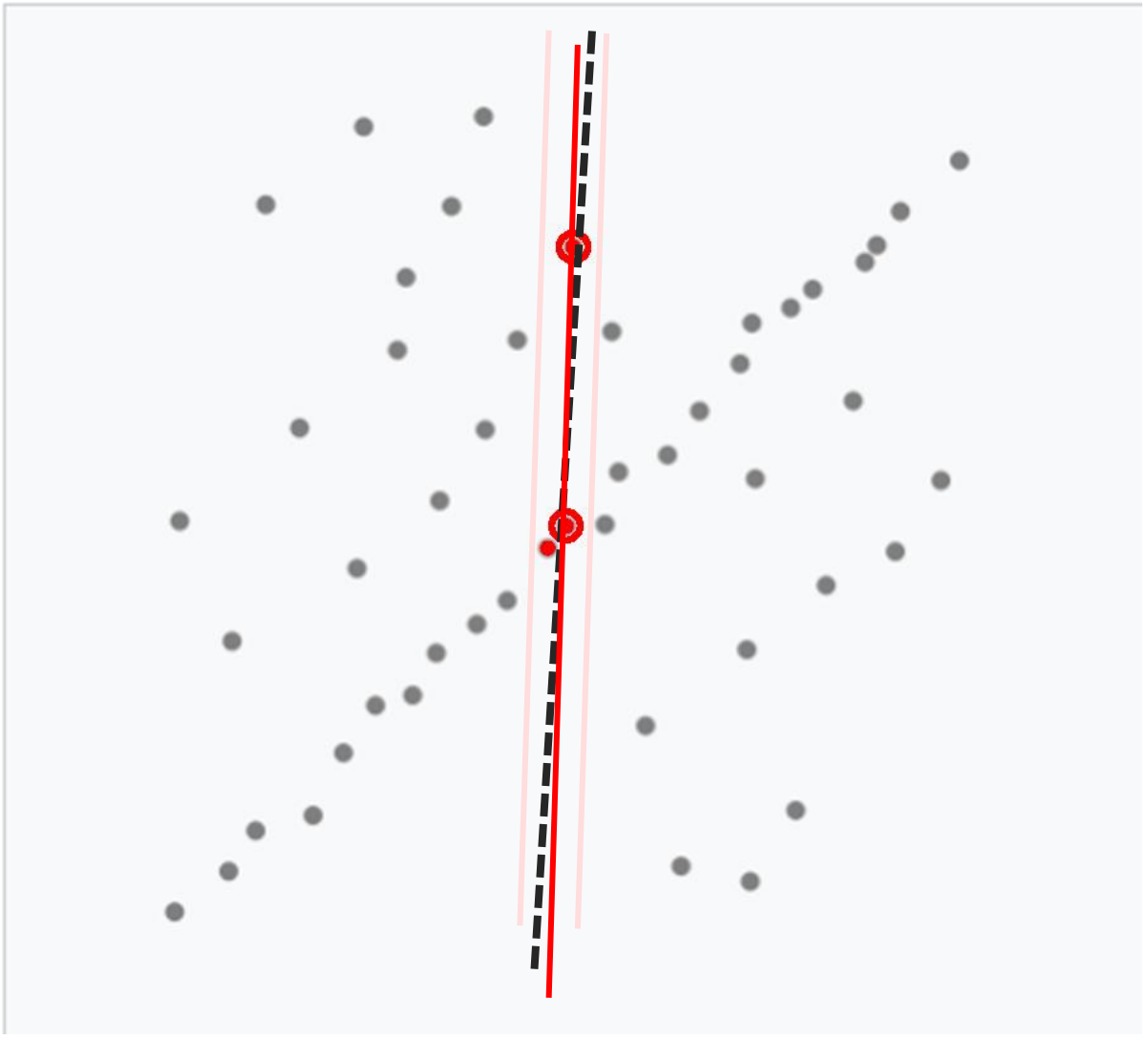
RANSAC – fitting a line

- Iteration 1



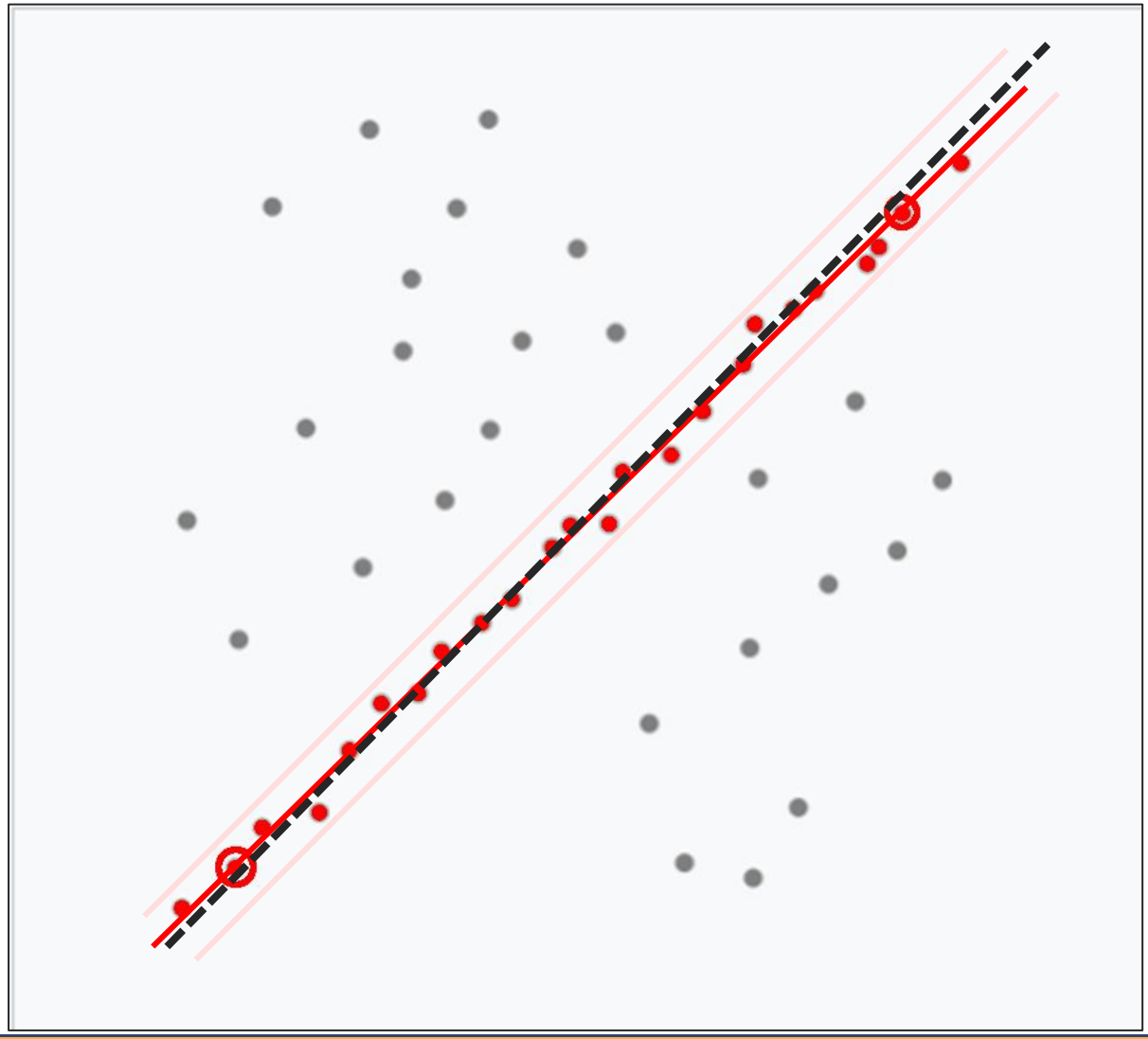
RANSAC – fitting a line

- Iteration … n



RANSAC – fitting a line

- Iteration … OK



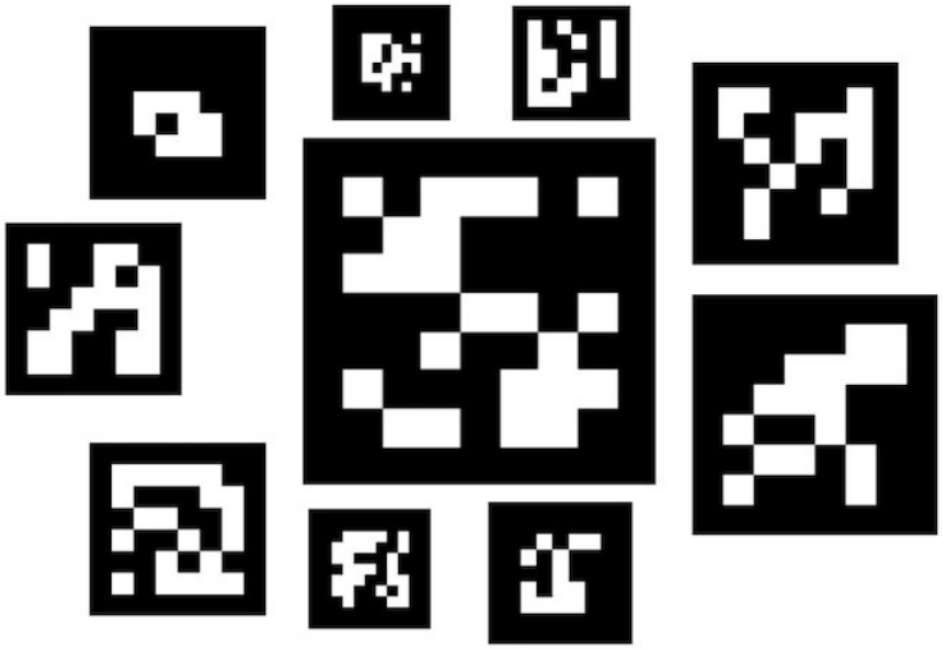
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AruCO Markers

- AruCO = Augmented Reality University of Cordoba

- Binary square fiducial markers
- Used for camera pose estimation
- Detection is robust, fast and simple.



AruCO - OpenCV

- OpenCV aruco module = detection of AruCO and tools for pose estimation and camera calibration.

- ChArUco combine ArUco markers with traditional chessboards to allow an easy and versatile corner detection.

Camera calibration OpenCV

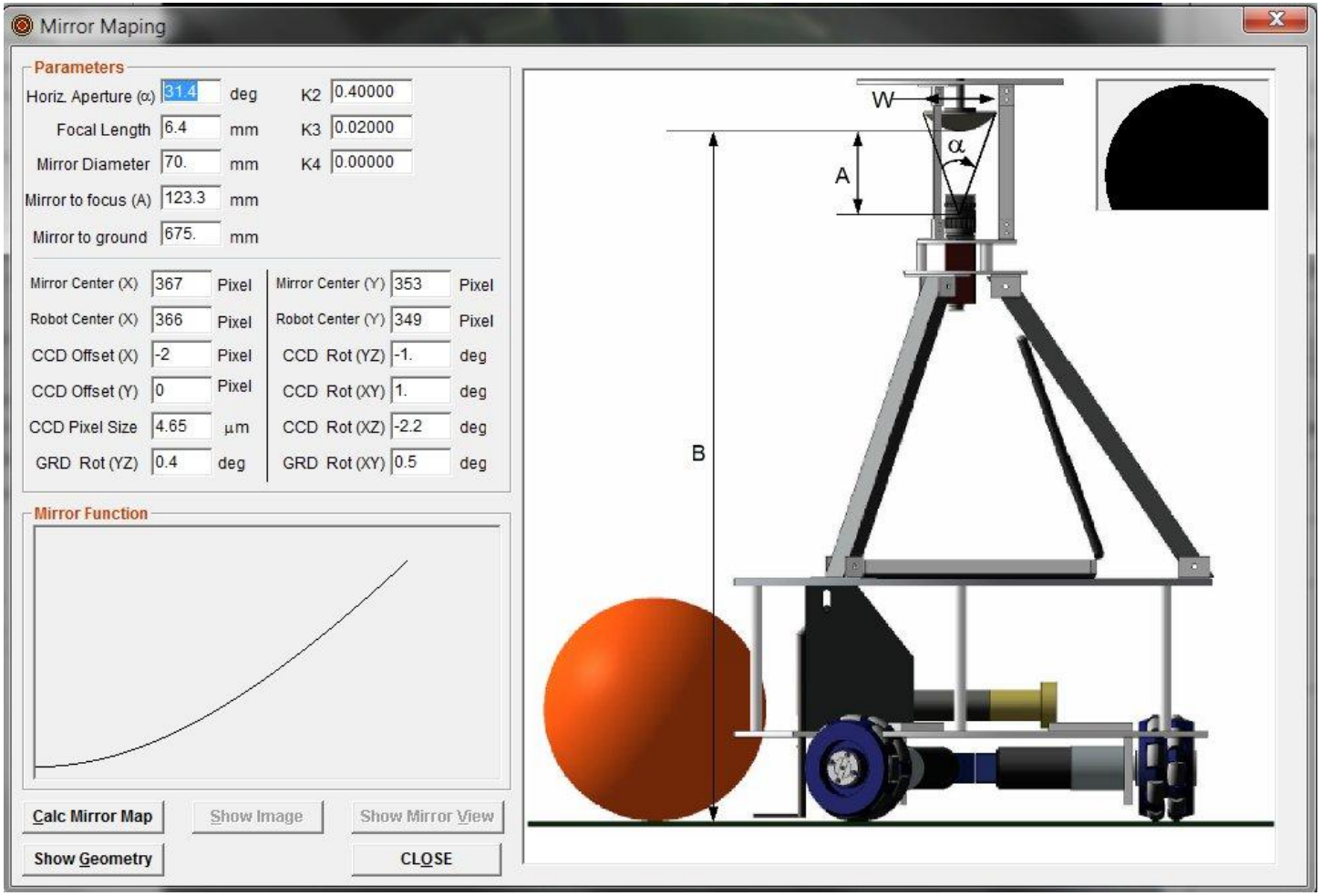
- cvFindChessboardCorners: internal corners of the chessboard
  - Optional: cvFindCornerSubPix: Refines the corner locations
- cvCalibrateCamera2: find intrinsic and extrinsic parameters from several views of chessboard.
- solvePnP: finds an object pose from 3D-2D point correspondences.

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- AruCO markers
- Example and Demos

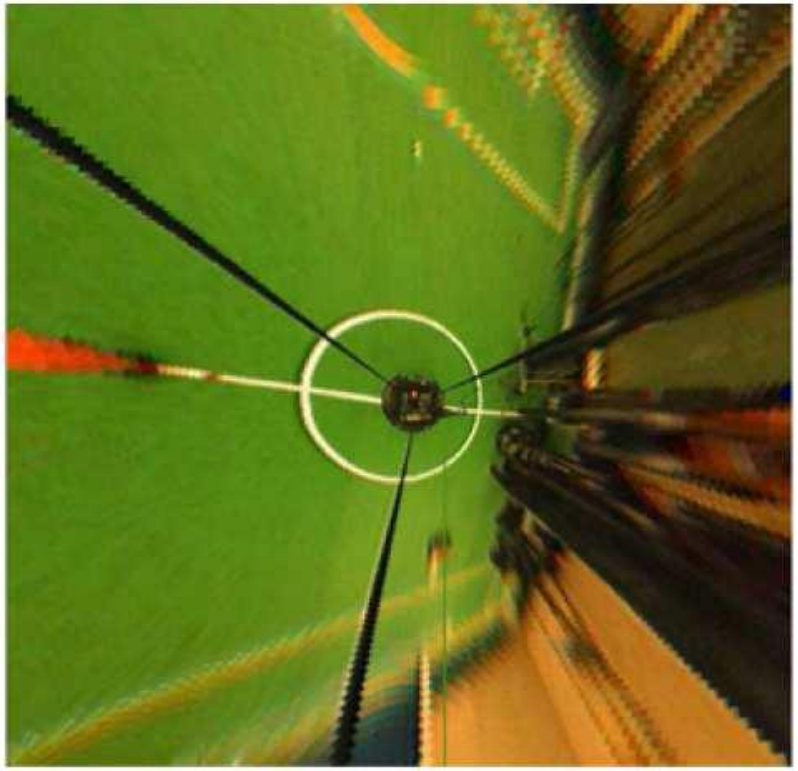
Vision System – example

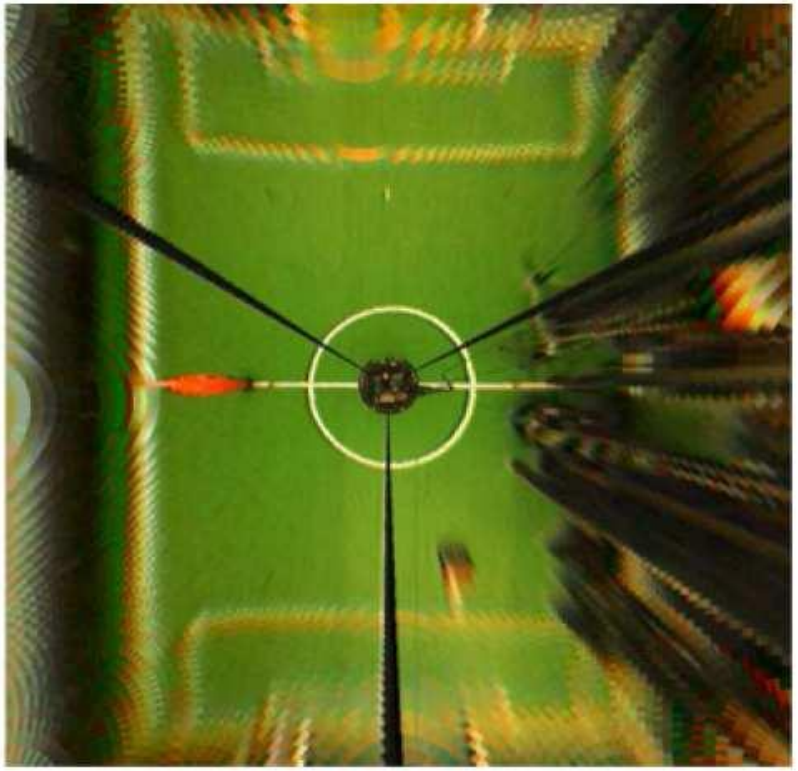
- A catadioptric system is a vision system based is based on one camera and one mirror system



Vision System – example

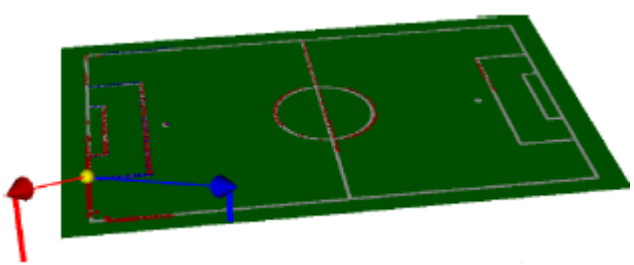
Such a system suffers from mechanical misalignments originating from several possible different reasons. By using the camera model, the catadioptric model and estimating the geometry parameters that introduce the distortion, one can correct the original image to obtain a fairly accurate distance map at the ground level

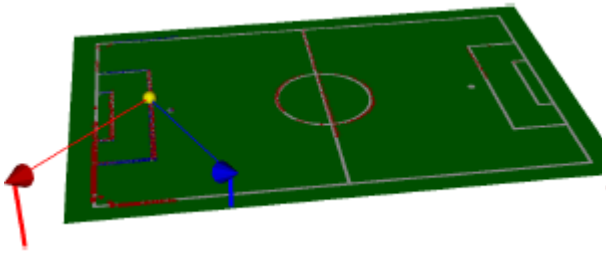


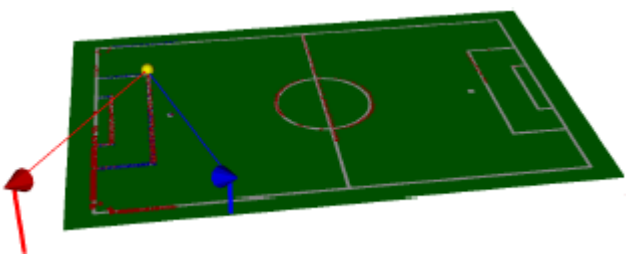


Vision System – example

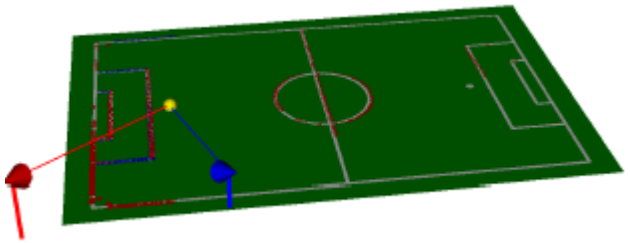
Ground truth for ball positions









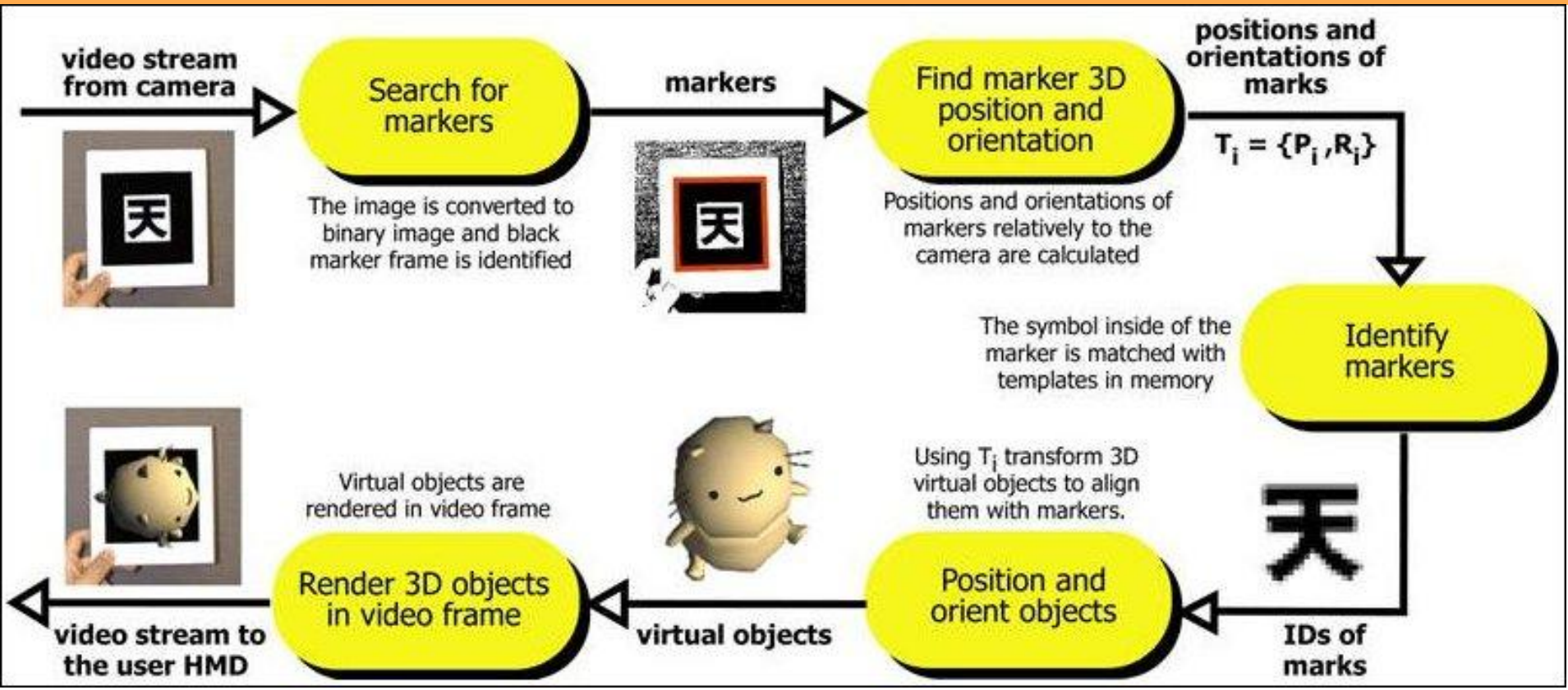


OPENCV CAMERA CALIBRATION DEMO

ARToolkit

- C/C++ library
- Estimates position/orientation of a camera
- Marker detection
- Camera calibration procedure
- Allow Real Time AR application
- 3D models reading
- Open-source

ARToolkit principle



- captures video of the real world
- searches through each video frame for any square shapes.
- If a square is found, mathematics to calculate the position of the camera relative to the square.
- graphics model is drawn from that same position.
- model is drawn on top of the video of the real world and so appears stuck on the square marker.
- final output is shown back in the display, user looks through the display and see graphics overlaid on the real world.

ARToolkit – Camera Calibration



- Default camera properties are contained in the camera parameter file camera_para.dat, that is read in each time an application is started.
- should be sufficient for a wide range of different cameras.
- However using a relatively simple camera calibration technique it is possible to generate a separate parameter file for the specific cameras that are being used (http://www.hitl.washington.edu/artoolkit/documentation/usercalibration.htm)
- if camera parameters are known then the video image can be warped to remove camera distortions.

Some references

- Gary Bradski and Adrian Kaehler. *Learning OpenCV: Computer Vision with the OpenCV Library*. O'Reilly, Cambridge, MA, 2008.
- Olivier Faugeras Three-dimensional computer vision: a geometric viewpoint. MIT Press Cambridge, MA, USA ©1993
- Szeliski, R. (2010). Computer Vision: Algorithms and Applications, Springer
- Richard Hartley and Andrew Zisserman (2003). Multiple View Geometry in Computer Vision. Cambridge University Press. pp. 155–157. ISBN 0-521-54051-8.
- Z. Zhang, "A flexible new technique for camera calibration'", IEEE Transactions on Pattern Analysis and Machine Intelligence, Vol.22, No.11, pages 1330–1334, 2000.